Representational Dynamics and the Yang–Mills Programme Conceptual Retrospective, Minimal Substrate, Aggressive Hypotheses, and Reward-Ranked Strategy

Research programme note

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Abstract

This note takes stock of a representation-aware constructive programme for four-dimensional Yang–Mills theory. The programme has produced exact local and finite-block results, counterexamples to scalar worst-history estimates, and conditional representation-to-spectrum theorems, but it has not proved physical one-scale basin entry, repeated renormalization-group control, continuum Yang–Mills theory, or a Hamiltonian mass gap. We argue that the missing object is a normalized, state-weighted, topologically complete transfer dynamics between scales. We formulate a minimal substrate as a region–scale network of operator systems, states, and completely positive maps; state ten aggressive hypotheses; and report executable first probes of each. Two probes materially sharpen the programme: coordinated multi-star constraints acquire zero-spill directions on the certified finite block, and center labels together with a Pfaffian charge fail to compose under addition of a common flux context. A reward-ranked strategy selects a vacuum-weighted, normalized, multichannel one-step kernel inequality as the primary target, with coordinated polymer cancellation as the leading backup. All continuum and spectral conclusions remain explicitly conditional.

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1 Where the programme stands

The programme has converged on a sharper diagnosis. The principal missing object is not another isolated local estimate. It is a controlled middle dynamics connecting ultraviolet gauge geometry to infrared spectral structure.

Programme principle 1 (Revised programme thesis)**.** *The constructive Yang–Mills problem, in the present architecture, is to build a gauge-covariant, positivity-preserving, topologically complete renormalization dynamics whose normalized defect transfer is uniformly contractive in physical units, and whose contraction is complete enough to reconstruct a nonzero Hamiltonian spectral gap.*

This is stronger than saying that infrared rigidity must merely be transported across representations. Representation changes are not passive: they forget information, alter the meaning of locality, and create lifting, normalization, or exceptional-set defects.

The programme presently resembles a bridge with a developed ultraviolet entrance, a conditional infrared endpoint, and a missing central span.

Ultraviolet and finite-block side. The package contains explicit SU(2) heat-kernel enclosures, quotient–fiber certificates, nonlinear local charts, an exact Fréchet-dispersion identity, dependent Gaussian calculations, boundary-frustration counterexamples, state-weighted transfer lemmas, and partial center and integer topology certificates. These are restricted or finite statements.

Infrared and spectral side. The package contains an abstract representation calculus and conditional theorems showing that a common temporal decay rate on a dense recovered observable core excludes spectrum below that rate. Countermodels show that observable completeness, positivity, rate uniformity, and the order of limits cannot be suppressed.

First missing physical implication. For every positive-mass retained history h, one needs a normalized comparison and chart-complement estimate of the form

$$\mu_h(A \cap G_h) \le e^{\ell_h} \nu_h(A \cap G_h), \tag{1}$$

$$\mu_h(G_h^c) \le q_{\mathrm{exc},h}, \tag{2}$$

such that

$$q_{\mathrm{low},h} + e^{\ell_h} q_{\mathrm{gauss},h} + q_{\mathrm{exc},h} \le q_{\mathrm{cont}} \tag{3}$$

uniformly in volume, cutoff, and admissible boundary state. The package closes this budget only in a centered reference model, not in the full interacting measure.

Claim ceiling. The strongest Yang–Mills-facing status remains C2: rigorous restricted-model theorems. The abstract transfer-to-spectrum statements are conditional. There is no physical one-scale basin theorem, repeated RG trajectory, continuum construction, Osterwalder–Schrader reconstruction, or Hamiltonian mass gap.

1.1 The material orientation correction

The orientation-correct Paper 05 result supersedes the exploratory Stage 44 zero-spill claim. The centered covariance spectrum is

$$\left\{0, \frac{239}{630}, \frac{239}{630}, \frac{33}{70}, \frac{33}{70}, \frac{33}{70}\right\}, \tag{4}$$

whereas the minimum external-curvature lift has strictly positive spectrum

$$\left\{\frac{37}{33}, \frac{37}{33}, \frac{37}{33}, \frac{391}{239}, \frac{391}{239}\right\}. \tag{5}$$

Every nonzero centered one-star displacement therefore spills curvature when lifted to link variables. The earlier zero mode resulted from centering raw signed plaquette rows before putting all six staples in a common rooted-link orientation. This correction is evidence that typed representation maps are part of the mathematics, not philosophical decoration.

2 The conceptual tensions

Tension	Lesson	Strategic response
Gauge invariance versus locality	Individual staples are covariant rather than invariant; their mutual dispersion is invariant.	Use local covariant coordinates together with invariant relational functionals.
Coarse-graining versus completeness	Erasing a boundary history can turn a valid weighted statement into a false worst-case statement.	Retain a conditionally sufficient boundary state and use positive transfer kernels.
Local repair versus realizability	An operation local in staple space can necessarily export curvature in link space.	Repair coordinated polymers or use normalized integration without local repair.
Energy versus entropy	An action barrier does not control normalized probability when multiplicity and denominators compensate it.	Prove a free-energy, relative-entropy, or partition-ratio inequality.
Typicality versus hostile boundaries	Rare but positive-mass histories can destroy a scalar conditional ceiling.	Use a state weight that transports the cost of entering and persisting in hostile states.
Continuous charts versus topology	Near-identity coordinates miss center flux, cut/orbit transitions, and integer charge.	Use a typed multichannel atlas and prove its closure under blocking.
Finite exactness versus physical uniformity	A certified finite-block spectrum need not survive volume, scale, or RG history.	Demand physical-unit uniformity along the tuned continuum trajectory.

Tension	Lesson	Strategic response
Euclidean decay versus Hamiltonian spectrum	A stochastic or finite transfer gap is not the Yang–Mills mass gap.	Preserve reflection positivity and a dense physical observable core through OS reconstruction.

2.1 What may and may not be compromised

The non-negotiable content is a state-separating gauge-invariant observable algebra, positivity sufficient for reconstruction, controlled locality in physical distance, topological completeness, cutoff and volume uniformity, and theory identity with nontrivial four-dimensional pure Yang–Mills.

The regulator, lattice action, microscopic gauge coordinates, a particular block map, pointwise worst-boundary bounds, single-star repair, and premature particle language may all be replaced. The programme should preserve phenomena and reconstruction certificates rather than a preferred set of coordinates.

3 Ontology and a minimal substrate

A physical universe is not taken to be a point in one preferred configuration space. It is a coherent family

$$\mathfrak{U} = \left\{(\mathcal{A}_R, \omega_R), \Phi_{R\to R'}\right\}_{R,R'} \tag{6}$$

of regional and scale-dependent observable systems, states, and admissible representation maps, equipped with compatibility, positivity, locality, covariance, limit, and reconstruction certificates.

Gauge fields are charts on this relational structure. Regulators are finite realizations. Renormalization maps express controlled forgetting. The continuum theory is a coherence class of compatible limiting realizations. Particles, when they exist, are stable spectral modes of the reconstructed dynamics.

3.1 The simplest adequate substrate

The simplest substrate that can realize all of these roles is a directed region–scale network of finite operator systems and completely positive maps. Let

$$\mathsf{RegionScale} = \{(O, \ell) : O \text{ is a region and } \ell \text{ is a scale}\} \tag{7}$$

be a poset. Attach an operator system $\mathcal{A}_{O,\ell}$ and a positive normalized state $\omega_{O,\ell}$ to each node. Regional inclusions are represented by unital $*$-embeddings. Coarse-graining observables is represented by unital completely positive maps; the dual maps reduce states and are completely positive and trace preserving.

In compact notation, exact coherence is a presheaf

$$\mathcal{U} : \mathsf{RegionScale}^{\mathrm{op}} \longrightarrow \mathsf{OpSys}_{\mathrm{ucp}}. \tag{8}$$

When finite representation errors must be retained, the appropriate object is a lax functor with controlled two-cell defects. For a composable triple, one records

$$\|\Phi_{02} - \Phi_{01}\Phi_{12}\|_{\mathcal{O}} \le \varepsilon_{012}(\mathcal{O}), \tag{9}$$

and proves that the errors remain summable on the declared observable class.

This substrate is preferable to taking Hilbert spaces as fundamental for the constructive stage. Positivity is native; irreversible coarse-graining need not be a homomorphism; commutative algebras reproduce classical probability; and Hilbert space can be reconstructed later by GNS or OS procedures. Gauge changes act as natural automorphisms, while physical observables lie in the appropriate invariant subobjects.

3.2 Finite coherence example

The executable supplement realizes the chain

$$M_2 \otimes M_2 \otimes M_2 \longrightarrow M_2 \otimes M_2 \longrightarrow M_2 \tag{10}$$

by successive partial traces on states and identity embeddings on observables. For the positive product state used in the certificate, the two-step and direct state reductions agree to 5.56×10^{-17}, and the fine/coarse observable pairing agrees exactly at machine precision. This is only a finite kinematic realization. It does not supply Yang–Mills dynamics, locality estimates, or a continuum limit.

4 Ten aggressive hypotheses

The hypotheses below are deliberately stronger than the proved package. They are useful only because each is paired with a mathematical signature and a falsification route.

Hypothesis 1 (Vacuum weight)**.** *The optimal positive retained-state weight is a coarse approximation to the vacuum boundary amplitude. A Doob transform by this weight converts hostile boundary histories into a contractive positive kernel.*

Hypothesis 2 (Physical Lyapunov exponent)**.** *On a complete nonvacuum observable core, the mass scale is the negative physical-length Lyapunov exponent of an OS-compatible representation cocycle:*

$$m = -\limsup_{L\to\infty} \frac{1}{L} \log \left\| \mathcal{R}_L \big|_{\mathbf{1}^\perp} \right\| . \tag{11}$$

This becomes a Hamiltonian mass only after positivity, completeness, and reconstruction are established.

Hypothesis 3 (Polymer boundary law)**.** *Although a one-star centered displacement has strictly positive spill, shared cochain and Bianchi constraints allow coordinated polymer displacements whose minimum external-curvature cost grows with $|\partial P|$ rather than $|P|$.*

Hypothesis 4 (Graded topology)**.** *The sufficient retained topological state is a graded cocycle containing continuous curvature, a center-valued two-cocycle, an integer flux lift, and the transition data needed for their composition.*

Hypothesis 5 (Free-energy rigidity)**.** *The transferable local rigidity functional is a normalized free-energy or conditional relative-entropy cost, not a bare action charge. Its useful feature is compatibility with data processing under coarse-graining.*

Hypothesis 6 (Representational curvature)**.** *Gauge fixing, orientation, charting, integration, and blocking define paths in a representation graph. The failure of alternative paths to commute is a quantitative curvature or holonomy defect whose accumulated norm must vanish or remain summable.*

Hypothesis 7 (Atlas-relative locality)**.** *No single coordinate presentation must simultaneously expose gauge invariance, microscopic realizability, positivity, and useful locality. Physical locality is a gluing property of overlapping covariant charts and regional observable systems.*

Hypothesis 8 (Counterexamples as completeness detectors)**.** *A hostile conditioning counterexample diagnoses information erased from the retained state. Iterative state refinement should terminate when the induced kernel becomes conditionally sufficient, or else demonstrate that the proposed finite state description is inadequate.*

Hypothesis 9 (Mass without a confinement premise)**.** *A mass gap can in principle be obtained from temporal rigidity of a complete local gauge-invariant observable algebra without assuming a Wilson-loop area law. Observable density and reflection positivity remain indispensable.*

Hypothesis 10 (Observable-selective recovery)**.** *Renormalization can be irreversible on microscopic coordinates while admitting exact or approximate recovery on the physical observable algebra. This gives an information-theoretic meaning to controlled forgetting without pretending that coarse-graining is invertible.*

5 Completed first probes

Every hypothesis received a formal definition, an executable finite or abstract probe, a hostile test, and a claim-level audit. The complete output is stored in `hypothesis_probe_results.json`. Twelve standard-library tests reproduce the calculations.

ID	Probe result	Status
H1	For $Q = \left(\begin{smallmatrix} 0 & q^2 \\ 1 & 0 \end{smallmatrix}\right)$, the positive weight $(q, 1)$ gives weighted ceiling $q = 0.08$, while the scalar row ceiling is one.	Finite positive-kernel mechanism survives; its identification with a Yang–Mills vacuum weight is open.
H2	Uniform factor 0.7 gives rate $-\log 0.7 = 0.356675$. The strict factors $e^{-1/n}$ have zero asymptotic physical rate.	Pointwise strict contraction is falsified as a sufficient premise.
H3	One star has no zero-spill modes. The central four-star and eight-star systems have respectively four and seven numerical zero-spill directions.	Collective cancellation is plausible, but no boundary scaling or nonlinear repair is proved.
H4	Two integer fluxes with identical center label and Pfaffian charge evolve to different charges after the same added context.	Center plus total charge is compositionally incomplete; a fuller integer carrier is necessary.
H5	The event identity $-\log \mu(A) = D(\mu(\cdot\|A)\|\mu)$ and data processing hold in the finite model. Charge five with multiplicity one thousand yields bad probability 0.8708.	Free-energy formulation survives; action-only suppression is falsified.

ID	Probe result	Status
H6	Centering and orientation have commutator rank two and operator norm one. The raw and oriented covariance spectra differ.	A real representation-path defect is present; the curvature interpretation remains conjectural.
H7	Five thousand deterministic quaternion samples reconstruct with error below 1.4×10^{-15} and radius below $\pi/2$; $-I$ is regular in the negative-center chart.	The two-center SU(2) atlas supports atlas-relative locality.
H8	A four-state regular/defect partition has lumpability defect one. Two refinement rounds isolate the four states and reduce the defect to zero.	A finite completeness-detector algorithm is viable.
H9	Spectrum $\{0, 0.4, 1.2\}$ gives true gap 0.4; an incomplete core seeing only the high mode reports 1.2.	The spectral mechanism survives, and density is shown indispensable.
H10	Coarse averaging and observable lifting recover the invariant two-dimensional algebra exactly while erasing a gauge-copy antisymmetric direction.	Observable-selective recovery exists in a toy model only.

5.1 The coordinated-spill signal

The multi-star calculation uses the oriented centered star map A_P and the external plaquette map B_P. For feasible star data y, it eliminates $\ker A_P$ and computes

$$\inf_{x:A_P x=y} \|B_P x\|^2. \tag{12}$$

The resulting numerical spectra are summarized below.

Polymer	rank A_P	zero modes	minimum positive	maximum
one star	5	0	1.121212	1.635983
adjacent pair	10	0	0.842105	2.181818
central four	18	4	0.591234	2.255172
all eight interior links	24	7	0.750000	1.428571

The one-star row reproduces the exact orientation-correct spectrum. The new zero modes show that collective cancellation exists on this finite complex. They do not show that arbitrary repair directions can be decomposed into such modes, that nonzero modes have boundary rather than volume cost, or that a finite nonlinear transformation has a controlled Jacobian.

5.2 A compositional topology obstruction

Let

$$n = (-1, -1, -1, -1, -1, -1), \tag{13}$$

$$n' = (-1, -1, -1, -1, 1, 1). \tag{14}$$

Both have center label $(1,1,1,1,1,1)$ modulo two and Pfaffian charge one. After adding the same context $e_{01} = (1,0,0,0,0,0)$, their charges become zero and two. Hence the pair "center label plus total Pfaffian charge" is not closed under contextual composition. Retaining the full six-component integer flux, or equivalent bilinear pairing data, is not optional in a blocking calculus.

6 Reward-ranked strategy

Reward is assessed qualitatively by four criteria: how early the target lies in the Clay implication chain, how many later modules reuse it, how decisively failure redirects the programme, and whether a finite certificate can probe it before a long analytic investment.

Rank	Target	Reward tier	Reason
1	Vacuum-weighted normalized multichannel one-step kernel	A+	It is the first open physical implication and composes H1, H4, H5, and H8. Success creates basin entry; failure identifies a missing state or normalization mechanism.
2	Exact coordinated-polymer spill scaling	A	New zero modes give a cheap, decisive test of the repair backup. An extensive lower bound would kill the route; a boundary law would create a new constructive mechanism.
3	Topology-complete retained state closed under blocking	A	H4 already falsifies a tempting smaller state. Closure is required before any multichannel kernel can be iterated.
4	Automated boundary-state sufficiency refinement	A–	H8 can turn counterexamples into a systematic retained-state discovery procedure and prevent repeated history-erasure errors.
5	Quantitative representation commuting-square calculus	B+	It catches orientation, chart, and limit-order defects but does not by itself provide probability suppression.
6	Atlas gluing and exceptional-set summability	B+	Necessary for a full measure, but best attacked after the normalized kernel and retained state are frozen.
7	Lyapunov-to-OS spectral theorem	B	High conceptual value and reusable, but it lies downstream of the present physical bottleneck.
8	Observable recovery/error-correction formulation	C	Potentially unifying, but presently a toy analogy rather than a source of the missing nonlinear estimate.

6.1 Primary mathematical target

Let Σ_n be the retained state at scale n, including continuous collar data, center flux, orbit/cut information, and a compositionally sufficient integer topological carrier. Let $D_{n,\alpha}(\sigma, d\sigma')$ be the normalized bad-extension kernel for channel α. The highest-reward theorem is to construct positive weights h_n and penalties λ_α satisfying

$$\boxed{\sup_{\sigma\in\Sigma_n} \frac{1}{h_n(\sigma)} \sum_\alpha e^{\lambda_\alpha} \int_{\Sigma_{n+1}} h_{n+1}(\sigma') D_{n,\alpha}(\sigma, d\sigma') \le q < q_{\mathrm{crit}}.} \tag{15}$$

The quantifiers must be uniform over positive-mass histories, finite volumes, and cutoffs on the tuned trajectory. The state weight must be constructed without using the desired clustering or gap.

6.2 Execution order

1. Freeze one SU(2) heat-kernel regulator, block, exposure order, and positive-mass retained-history space.
2. Prove conditional sufficiency of the retained state. Use refinement counterexamples to expose missing center, continuous, orbit, or integer data.
3. Construct the exact normalized positive kernel and solve its finite Perron problem. Test whether the positive eigenweight behaves like a coarse vacuum amplitude.
4. Prove the boundary-uniform continuous good-channel comparison, including all normalizers and chart complements.
5. Prove charged numerator inequalities for center, cut/orbit, and full integer topology channels.
6. Compose the kernel rows and verify a strict connected-set threshold.
7. Prove stability of the state space, weight, positivity, and constants under one further block step before attempting indefinite iteration.
8. Only after repeated physical-unit contraction, transport a complete observable core and invoke the conditional OS spectral theorem.

The coordinated-polymer route remains a backup. Its next decisive question is whether the nonzero minimum-spill cost of a growing polymer is bounded by a constant times its boundary. If the cost remains extensive, the repair route should be abandoned in favor of direct normalized integration.

7 Limits and concluding reformulation

The operator-system substrate supplies a clean kinematic language, not a Yang–Mills construction. Completely positive maps do not automatically encode relativistic locality, reflection positivity, asymptotic freedom, topological completeness, or theory identity. Those remain certificates to be proved for the selected regulator and limit.

Likewise, the multi-star zero modes are not a polymer theorem; the topology collision is not a topological sector-weight estimate; relative-entropy identities are not quasi-factorization; finite

partition refinement is not a proof that a finite sufficient Yang–Mills boundary state exists; and a toy OS semigroup is not a continuum reconstruction.

The programme's strongest conceptual compression is therefore

> The Yang–Mills mass gap is stable loss of nonvacuum relational information under physical scale evolution, reconstructed as spectral rigidity.

This is an interpretation with a precise proposed mathematical carrier: a complete, positive, normalized, multichannel transfer cocycle whose nonvacuum sector is uniformly contractive in physical units. Establishing that carrier for four-dimensional pure Yang–Mills remains the central open task.